

Study Questions: The Invention Roadmap

Analytical Questions

1. Active Inference frames planning as expected free energy minimization. Explain the two components of expected free energy (pragmatic value and epistemic value) and how they apply to an inventor choosing between two projects: one with high commercial potential but major technical uncertainty, and one with modest commercial potential but well-understood technology.
2. Staged development defers irreversible decisions until uncertainty is reduced. Using a specific invention example, explain why investing heavily in manufacturing before validating the product concept is a violation of expected free energy minimization. What is the excess free energy created by skipping the concept validation stage?
3. Compare the planning strategies of SpaceX (staged development from Falcon 1 to Falcon 9 to reusable operations) and a hypothetical competitor that tried to build a fully reusable rocket from day one. Using Active Inference concepts, explain why SpaceX's staged approach was more likely to succeed.
4. The concept of option value suggests that keeping multiple approaches open (even at some cost) can be preferable to committing to one approach prematurely. Under what conditions does option value outweigh the cost of maintaining multiple paths? How does the Active Inference framework formalize this trade-off?
5. Percy Spencer's microwave oven took 20 years from discovery to consumer adoption, with multiple plan revisions along the way. Analyze the roadmap of this invention using the staged development framework. Where did structure learning events force plan revisions? Where could earlier planning have anticipated the obstacles?
6. The concept of planning horizon suggests that detailed planning should not extend beyond the range of the generative model's useful predictions. How does an inventor determine their planning horizon? What signs indicate that a plan is extending beyond the horizon into unproductive speculation?

7. Explain how learning from earlier stages should update the plan for later stages. Use a concrete invention example where Stage 2 results required significant revision of the Stage 3 plan. What information from Stage 2 made the original Stage 3 plan obsolete?
8. Resource allocation under uncertainty follows the principle: invest where expected free energy is highest. Early in a project, fundamental viability questions dominate. Later, operational questions dominate. Describe how the resource allocation for a medical device invention should shift across five stages, from problem identification to regulatory approval.
9. The sunk cost fallacy causes inventors to continue with a failing plan because they have already invested resources. How does the Active Inference framework handle sunk costs? Why are sunk costs irrelevant to expected free energy calculations? What cognitive practices can help inventors overcome the sunk cost bias?
10. Adaptive roadmaps include explicit pivot criteria — conditions under which the plan fundamentally changes. Design three pivot criteria for an inventor developing a new educational technology platform. Each criterion should specify: the evidence that would trigger the pivot, the nature of the pivot, and why this particular evidence is decisive.

Applied Questions

1. Assess your invention project's current stage. What is the single highest-uncertainty question remaining? Design a plan to resolve this uncertainty within the next two weeks, specifying actions, evidence thresholds, and decision criteria.
2. Build a decision tree for the next critical decision in your invention project. Identify at least three branches (go, modify, pivot). For each branch, specify the evidence that would trigger it and the subsequent action plan.
3. Rate each action in your current plan on both epistemic value (uncertainty reduction) and pragmatic value (goal advancement). Which action has the highest combined value? Are you doing that action first? If not, why?
4. Examine your plan for premature optimization — places where you are investing resources in aspects that will only matter if more fundamental questions are answered

affirmatively. Identify one instance and describe how you would reallocate those resources.

5. Write your pivot criteria now, while you are still relatively detached from the project. Under what specific conditions would you: (a) continue with the current approach, (b) modify the approach significantly, or (c) abandon the project entirely? Be specific and measurable.
6. Calculate the planning horizon for your invention project. How many stages ahead can you plan with useful detail? Beyond that, what is the appropriate level of planning (directional intent, rough goals, or nothing at all)?
7. Consider the opportunity cost of your invention project. Every month you spend on this project is a month not spent on other projects. How does expected free energy analysis help you decide whether to continue, pivot, or start something new?
8. Design a one-month sprint plan for your invention that maximizes information gain per unit of time invested. What are the three most informative experiments or observations you could conduct, and in what order should you conduct them?
9. Imagine your invention project two years from now, from two different perspectives: (a) the version where everything goes well, and (b) the version where the project had to pivot significantly. What would the "everything goes well" plan look like? What would the pivot look like? What decision points separate these two futures?
10. Create a one-page "roadmap on a napkin" for your invention — the minimum viable plan that captures: current state, next milestone, key decision point, biggest risk, and first action. This should be simple enough to explain to a friend in five minutes and specific enough to guide your next week of work.